

constraints on $u, v \leftrightarrow X, Y$

Suppose the joint probability mass function of random variables X and Y is $p_{X,Y}(u,v) = 2^{-(u+v)}$ whenever u and v are both positive integers, and is zero otherwise. What is the probability that $Y + (X - 2)^2$ is greater than or equal to two?

- (a) 7/8
- (b) 1/2
- (c) 1/4
- (d) 1/8
- (e) 1/16
- (f) 3/4
- (g) 15/16
- (h) 3/8
- (i) 3/16
- (j) 5/16
- (k) 7/16
- (l) None of these

$$p_{X,Y}(u,v) = 2^{-(u+v)}$$

$$P(Y + (X-2)^2 \geq 2)$$

$$P(Y + (X-2)^2 < 2)$$

$$X, Y \in \mathbb{Z}^+$$

$$\rightarrow P(Y + (X-2)^2 = 1)$$

$$Y = 1 - (X-2)^2$$

$$1 \leq 1 - (X-2)^2$$

$$(X-2)^2 \leq 0 \quad \text{but } (X-2)^2 \geq 0$$

$$\text{so } (X-2)^2 = 0$$

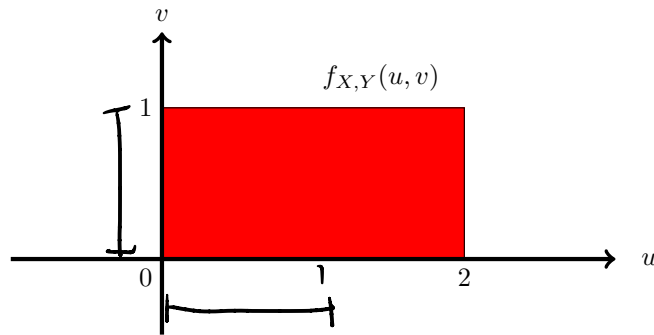
$$\rightarrow X = 2$$

$$Y = 1 - 0^2$$

$$p_{X,Y}(2,1) = 2^{-(2+1)} = \frac{1}{8}$$

$$p_{X,Y}(2,1) = \frac{1}{8}$$

If random variables X and Y have a joint pdf equal to $f_{X,Y}(u,v) = uv$ in the red rectangle in the figure below and which is zero elsewhere, then what is the probability that X is less than one?



$P(X < 1)$?

- (a) $1/4$
- (b) $1/2$
- (c) $1/8$
- (d) $1/16$
- (e) $1/3$
- (f) $2/3$
- (g) $3/4$
- (h) $3/8$
- (i) $3/16$
- (j) $5/16$
- (k) $5/8$
- (l) None of these

$$\begin{aligned}
 P(X < 1) &= \int_0^1 \int_0^1 uv \, du \, dv \\
 &= \frac{1}{2} \int_0^1 [vu^2]_0^1 \, dv \\
 &= \frac{1}{2} \int_0^1 v \, dv \\
 &= \frac{1}{4} [v^2]_0^1 = \frac{1}{4}
 \end{aligned}$$

If X is a random variable that is uniform on $[0, 3]$ and $Y = e^X$, then what is the pdf value $f_Y(8)$?

Uniformly distributed r.v. on an interval (a, b) has
pdf

$$f_X(u) = \begin{cases} \frac{1}{b-a} & , a < u < b \\ 0 & , \text{else} \end{cases}$$

(a) $1/24$

(b) $1/8$

(c) $1/3$

(d) $1/4$

(e) $1/12$

(f) $(\ln 8)/8$

(g) $(3 \ln 8)/8$

(h) $(\ln 8)/24$

(i) e^8

(j) $\ln 8$

(k) e^3

(l) None of these

$$f_X(u) = \frac{1}{3}$$

$$Y = e^X \rightarrow Y = [1, e^3]$$

$$\rightarrow X = \ln(Y)$$

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|$$

$$f_Y(y) = f_X(\ln(y)) \cdot \left| \frac{1}{y} \right|$$

$$f_Y(8) = f_X(\ln(8)) \cdot \frac{1}{8}$$

$$= \frac{1}{24}$$